

Curriculum Vitae

Name	Bhavan Chithambaram
Portfolio	https://bhavanc.github.io/portfolio/
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Educations

Oct 2025 – Present	Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau Master of Science (M.Sc.) – Commercial Vehicle Technology
Aug 2019 - Apr 2023	Sri Sairam Engineering College, Chennai Bachelor of Engineering (B.E) - Mechanical Engineering Grade: 8.76 Thesis: Developed an ACC system with vehicle detection and lane detection. Created a sensor fusion algorithm to identify the Most Important Object (MIO). Control over the vehicle's longitudinal movement was designed in Stateflow with sensor data. Built a feedback model for vehicle dynamics with a controller for lateral movements. Tested and validated the system in a custom virtual environment using MATLAB & Simulink.

Work Experiences

Aug 2024 – Jul 2025	Engineer – Product Development Raptee.HV, Chennai - Modelled Battery pack and performed various simulations – MATLAB & Simulink. - Developed an algorithm for BMS's Passive balancing – Reduced heat dissipation (33) %. - Created different charging strategies to reduce charging time and increase battery life. - Performed Design & Validation Plan for Battery pack – Testing and Validation.
Jul 2023 – Aug 2024	GET - Vehicle Engineer - Created algorithms in MATLAB for VCU - Throttle mapping and Range Estimation. - Built a MiL in Simulink and Stateflow for simulation - Power train, Battery and Motor. - Testing, and Validation - Control logic, Vehicle drive data and Startup sequences. - Designed a Digital twin Simulink model to analyse Vehicle performance using CAN data.
Jan 2021 – Feb 2022	Suspension Engineer Team Evega Racing SAEINDIA eBaja 2022, Chennai - Determined the optimum suspension geometry - Vehicle dynamics and simulations. - Designed system components - Withstanding all the dynamic loads in the operating condition. - Performed FEA studies using ANSYS for Structural and Thermal analysis. - Modelled a power train of an Electric ATV vehicle in Simulink. - Simulated and validated both the Motor and battery pack.

Aug 2021 – Sep 2021

Model-Based Design of Automotive Systems Intern

Decibels Lab Pvt. Ltd, Bangalore

- Modelled and simulated a vehicle dynamic body - KTM bike with an Antilock Braking System.
- Modelled and simulated a Nissan leaf in the Scilab-Xcos – HVAC system.

May 2021 - Jul 2021

Research Intern

Genik Technologies Pvt. Ltd, Chennai

- Created an **algorithm** for a Traffic Signal Simulator - Machine Learning (**Python**).
- Detected the traffic in a certain area - Sensor (Camera) and **Computer vision technology**.
- Validated the algorithm with different inputs that change the traffic signal - Based on data from the camera.

Projects

Oct 2022 – Dec 2022

Field-Oriented Control in Permanent Magnet Synchronous Motor.

- Improved the response of Field Oriented Control in PMS Motor - Reinforcement Learning.
- **Reinforcement Learning as a Controller** - Trained under various specifications.
- Created an algorithm for training - based on a reward function.
- The trained agent was proven to respond faster than PI controllers.

Feb 2022 – Apr 2022

Lattice Optimization on a Steering Knuckle.

- Optimized a Steering Knuckle in its lattice structure - nTopology software.
- Performed multiple advancements in a component - **Stress distribution** and Heat dissipation.
- Weight reduction of a component was achieved - A total **reduction of 22%**.

Publications

Jan 2024

Modelling, Simulation and Virtual Validation of Adaptive Cruise Control (ACC) Algorithm based on Sensor Fusion. [Link](#)

2023 International Conference on Science, Engineering and Technology Practices for Sustainable Development (ICSETPSD). **EAI**.

Feb 2023

Improvised FOC and Speed Estimation of PMSM using Reinforcement Learning and Neural Networks. [Link](#)

2022 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS). **IEEE Xplore**. Best paper award.

Languages

German – A2

English – C1